

# Pranav Roy

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## Summary

BBACA graduate with hands-on experience across 15+ published data and software projects, spanning statistical experimentation, machine learning, business intelligence, and full-stack development. Comfortable moving between advanced SQL, Python analytics, and interactive dashboarding to turn raw data into decisions stakeholders can act on. Seeking entry-level roles in Data Analytics, Data Science, or Software Engineering in Pune.

## Technical Skills

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|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>              | Python, SQL, JavaScript, Java                                                                                             |
| <b>Data &amp; Analytics:</b>   | Pandas, NumPy, scikit-learn, SciPy, Power BI (DAX, Power Query), Excel (Pivot Tables, Advanced Formulas)                  |
| <b>Statistical Methods:</b>    | Chi-square testing, t-test, logistic regression, ARIMA forecasting, K-Means clustering, A/B testing, confidence intervals |
| <b>Databases:</b>              | PostgreSQL, SQLite – CTEs, window functions, RANK/DENSE_RANK, joins, query optimization                                   |
| <b>Tools &amp; Deployment:</b> | Git/GitHub, Streamlit, Jupyter, VS Code, Tkinter                                                                          |

## Experience

### CodeAlpha – Artificial Intelligence Intern

2026

*Remote*

- Built three functional applications in a single internship cycle: TITAN, a real-time multi-object detection system (YOLOv8 + ByteTrack + OpenCV) sustaining 30+ FPS on CPU-only hardware; ARIA, a TF-IDF retrieval chatbot with LLM API fallback; and a 20-language desktop translator (Tkinter).
- Debugged model inference latency, API timeout failures, and GUI threading conflicts, documenting resolutions for reproducible deployment.

### CodeAlpha – Data Analytics Intern

2025

*Remote*

- Delivered an end-to-end sentiment analysis pipeline on 50,000 IMDB reviews: web scraping (BeautifulSoup), EDA with Pandas, TextBlob polarity scoring, and Matplotlib/Seaborn visualization in a reproducible Jupyter notebook.

### Forage Job Simulations – Tata iQ, Deloitte, AWS

2025–2026

*Remote*

- Performed EDA and predictive modelling for a credit card delinquency reduction case (Tata iQ) and built an interactive stakeholder dashboard.
- Designed a cloud architecture (EC2, S3, RDS) with data-flow diagrams for a forensic technology use case (Deloitte/AWS track).

## Education

### Modern College of Arts, Science and Commerce, Pune

Jul 2023 – Apr 2026

*Bachelor of Business Administration in Computer Applications (BBACA)*

## Projects

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### Bank Customer Churn & Retention Analytics

2026

*Python, scikit-learn, Power BI, Streamlit* | [GitHub](#) | [Live App](#)

- Built an interpretable logistic regression model on 10,000 customer records to predict individual churn risk, achieving 80.7% accuracy while explicitly reporting recall (17%) to account for class imbalance rather than relying on accuracy alone.
- Identified inactive single-product customers as the highest-risk segment (36.65% churn vs. 20.37% overall, 2.82x higher odds) using chi-square testing (Geography:  $\chi^2 = 301.26$ ,  $p < 0.001$ ) and logistic regression coefficients.
- Deployed a live Streamlit application enabling per-customer churn probability lookup, risk banding, and top-2 driver explanation; paired with a Power BI dashboard for stakeholder-facing segment analysis.

### Growth Experimentation & Retention Analytics

2026

*Python, A/B Testing, Streamlit* | [GitHub](#) | [Live App](#)

- Ran a chi-square A/B test on 120,000 sessions comparing an ad treatment against a PSA control, measuring a statistically significant lift from 1.79% to 2.55% conversion ( $p < 0.001$ ).
- Applied a pre-defined minimum detectable effect (MDE) threshold to the 95% confidence interval and recommended against full rollout despite a 43% relative lift, demonstrating the distinction between statistical and practical significance.
- Built a 4-stage funnel (Page View to Purchase) and cohort retention analysis by acquisition source, deployed as an interactive Streamlit dashboard with date-range and channel filters.

### Zepto vs. Blinkit – Quick Commerce Intelligence War Room

2026

*Python, PostgreSQL, Power BI, Excel* | [GitHub](#)

- Built an end-to-end pipeline across 4 source datasets applying t-tests, chi-square tests, K-Means clustering, and 30-day ARIMA demand forecasting.
- Delivered a 7-zone interactive Power BI dashboard (CEO summary, RFM segmentation, ARIMA forecast) using custom DAX measures and Power Query ETL.
- Identified and resolved 3 pipeline integrity bugs (cluster label inversion, narrative mismatch, statistical regeneration) through systematic root-cause analysis.

### Global Data Science Job Market Analysis

2026

*PostgreSQL, Pure SQL* | [GitHub](#)

- Analyzed 3,925 global data science job listings using advanced SQL – CTEs and window functions – to surface salary trends by country and experience level.
- Ranked the top 10 in-demand skills by posting frequency using `RANK()` and `DENSE_RANK()`, and quantified remote vs. on-site demand shifts year-over-year.

### Library Management System

2026

*Java, JUnit 5, Design Patterns* | [GitHub](#)

- Designed a clean-architecture Java application applying three GoF design patterns (Factory, Strategy, Singleton) to decouple business logic from data access.
- Achieved 17/17 passing tests in a JUnit 5 test suite, validating core lending, inventory, and member-management logic.

## Certifications

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- **Data Analysis with Python** – freeCodeCamp
- **Full SQL Queries & Data Science** – Simplilearn
- **Full MS Excel for Data Analysis** – Simplilearn
- **Career Essentials in Generative AI** – LinkedIn Learning / Microsoft